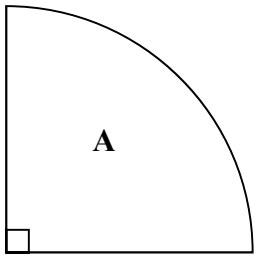
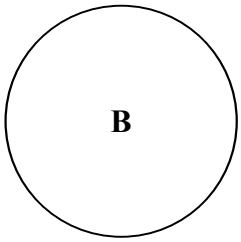
**30** **A** is in the shape of a quarter circle of radius 15 cm.
**B** is in the shape of a circle.




15 cm




The area of **A** is 9 times the area of **B**.

Show that the radius of **B** is 2.5 cm.

Area A = $(\pi \times 15^2) \div 4 = 56.25\pi$ cm$^2$

A is 9 times B, so area B = $56.25\pi \div 9$

Area B = $6.25\pi$ cm$^2$ = $\pi \times 2.5^2$

Hence radius B = 2.5 cm, as required.

**3 marks total**
Area of quarter circle
Area B = 6.25π
Radius 2.5 cm

**(Total for Question 30 is 3 marks)**